Dawon Park, Ph.D.

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Employment History

2026 – · · · ·

InnoCORE Fellow (Postdoctoral Fellow)
Korea Advanced Institute of Science and Technology (KAIST).
Department of Civil and Environmental Engineering.
T-RAX (Transformative Railway with AX) Research Center, InnoCORE Program.
Advisor: Prof. Jung-Wuk Hong.

Education

2021 – 2026

Ph.D.
Korea Advanced Institute of Science and Technology (KAIST).
Department of Civil and Environmental Engineering.
Thesis title: *Numerical Study of Blast Responses in an RC Slab Using Continuous Surface Cap Model.*
Advisor: Prof. Jung-Wuk Hong.

2019 – 2021

M.S.
Korea Advanced Institute of Science and Technology (KAIST).
Department of Civil and Environmental Engineering.
Thesis title: *Numerical Analysis on Seismic Behavior of a Three-story Shear Wall Reinforced Concrete Structure.*
Advisor: Prof. Jung-Wuk Hong.

2012 – 2019

B.S.
Pohang University of Science and Technology (POSTECH).
Department of Mechanical Engineering.
Military Service: Republic of Korea Marine Corps, 2015 – 2017.
Advisors: Prof. Junsang Doh and Prof. HangJin Jo.

Academic Contributions

Journal publications

24 **D. Park** and J.-W. Hong, "Seismic fragility of an L-shaped unreinforced masonry wall via incremental dynamic analysis and continuous damage classification," *Engineering Structures*, In revision. JCR Q1, 92.0% IF 7.6

23 **D. Park**, H. Kim, and J.-W. Hong, "Explainable AI analysis of multi-stage damage descriptors for debris flow impact on a transmission tower," *Engineering Applications of Artificial Intelligence*, In revision. JCR Q1, 96.9% IF 9.0

22 **D. Park**, M. Kim, S. Kim, H. Lee, S. Jin, and J.-W. Hong, "Multi-metric optimization of Mohr–Coulomb parameters in SPH granular flow simulation," *Engineering Geology*, 2026, Accepted. JCR Q1, 95.9% IF 10.3

21 T. Lee, Y. Lee, **D. Park**, Y. Choi, and J.-W. Hong, "ALE simulation and machine learning surrogate modeling of airborne blast loading from cuboid explosives," *Advances in Engineering Software*, vol. 223, p. 104 315, 2026. JCR Q1, 91.3% IF 6.3

20 **D. Park** and J.-W. Hong, "Sensitivity of blast responses in a reinforced concrete slab to damage evolution parameters," *Computational Mechanics*, 2026. JCR Q1, 92.3% IF 4.5

19 **D. Park**, H. Kim, M. Kim, and J.-W. Hong, "AI surrogate model for debris flow impact on a transmission tower using SPH–FEM simulations," *Computers & Structures*, vol. 328, p. 108 260, 2026. JCR Q1, 85.8% IF 5.6

18 **D. Park**, S. Kim, H. Lee, and J.-W. Hong, "Experimental assessment of the dynamic characteristics of masonry walls under incremental shake-table loading," *Journal of Building Engineering*, vol. 125, p. 116 165, 2026. JCR Q1, 93.0% IF 8.1

17 **D. Park**, Y. Kim, H. Lee, and J.-W. Hong, "Novel detonation-location mapping method for blast damage assessment with application to a 2D concrete slab," *Structures*, vol. 94, p. 113 216, 2026. JCR Q1, 81.6% IF 4.9

16 **D. Park**, T. Lee, Y. Lee, and J.-W. Hong, "Numerical study of RC slab failure mechanisms under varying blast loads," *Journal of Building Engineering*, vol. 120, p. 115 435, 2026. JCR Q1, 93.0% IF 8.1

15 Y. Choi, **D. Park**, S. Kim, and J.-W. Hong, "Geometric limits for the seismic design of unreinforced masonry fence walls," *Earthquake Engineering & Structural Dynamics*, vol. 55, no. 1, pp. 130–149, 2025. JCR Q1, 81.6% IF 4.9

14 **D. Park**, Y. Hwang, S. Jin, and J.-W. Hong, "Optimization of rate-dependent material parameters for accurate blast response prediction in reinforced concrete slabs," *Computers & Structures*, vol. 318, p. 107 961, 2025. JCR Q1, 85.8% IF 5.6

13 **D. Park**, W. Jang, E. Cheon, and J.-W. Hong, "Failure analysis of a transmission tower by Non-Newtonian debris flow impact with user-defined SPH particles and finite elements," *Results in Engineering*, vol. 28, p. 107 183, 2025. JCR Q1, 97.5% IF 9.4

12 **D. Park**, M. Kim, and J.-W. Hong, "Sequential failure-based fragility assessment of a transmission tower under debris flows," *Engineering Structures*, vol. 343, p. 121 097, 2025. JCR Q1, 92.0% IF 7.6

11 **D. Park**, S. Kim, S. Jin, Y. Choi, H. Kim, and J.-W. Hong, "Seismic vulnerability assessment of leaning U-shaped masonry fence walls," *Earthquake Spectra*, vol. 41, no. 3, pp. 2177–2202, 2025. JCR Q2, 73.8% IF 3.9

10 H. Kim, Y. Choi, **D. Park**, M. Lee, G. Lee, B. Kim, and J.-W. Hong, "Numerical evaluation of acoustic emission (AE) sensors by lead-zirconate-titanate (PZT) geometric design," *Sensors and Actuators A: Physical*, vol. 366, p. 115 031, 2024. JCR Q1, 82.9% IF 4.9

9 **D. Park**, T. Lee, Y. Lee, Y. Choi, and J.-W. Hong, "Blast simulations of a reinforced concrete slab using the continuous surface cap model (CSCM)," *Journal of Building Engineering*, vol. 96, p. 110 603, 2024. JCR Q1, 94.8% IF 7.4

8 Y. Choi, **D. Park**, S. Kim, and J.-W. Hong, "Investigation on the behavior factor of leaning masonry wall structures," *Bulletin of Earthquake Engineering*, vol. 21, pp. 2987–3009, 2023. JCR Q1, 80.5% IF 3.8

7 Y. Choi, **D. Park**, S. Kim, S. Jin, and J.-W. Hong, "Experimental and numerical analyses of seismic behaviors of L-shaped unreinforced masonry walls," *Engineering Structures*, vol. 297, p. 116 989, 2023. JCR Q1, 90.4% IF 5.6

6 Y. Lee, T. Lee, **D. Park**, Y. Choi, and J.-W. Hong, "Optimal mesh size in three-dimensional arbitrary Lagrangian-Eulerian method of free-air explosions," *Journal of Computational Structural Engineering Institute of Korea*, 2023, KCI.

5 H. Chen, D. Cho, K. Ko, C. Qin, M. Kim, H. Zhang, J. Lee, E. Kim, **D. Park**, X. Shen, J. Yang, H. Ko, J.-W. Hong, J. Kim, and S. Jeon, "Interdigitated 3D heterogeneous nanocomposites for high-performance mechanochromic smart membranes," *ACS Nano*, vol. 16, pp. 68–77, 2022. JCR Q1, 94.3% IF 17.1

4. Y. Choi, **D. Park**, S. Kim, and J.-W. Hong, "Seismic performance of crack-damaged masonry wall structures via shaking table tests," *Structures*, vol. 45, pp. 2272–2291, 2022. JCR Q2, 73.0% IF 4.1

3. **D. Park**, Y. Choi, S. Kim, and J.-W. Hong, "Failure analysis of a U-shaped concrete-block masonry wall structure via shaking table tests," *Engineering Failure Analysis*, vol. 142, p. 106 832, 2022. JCR Q1, 79.7% IF 4.0

2. G. Bae, G. Choi, C. Ahn, S. Kim, W. Kim, Y. Choi, **D. Park**, D. Jang, J.-W. Hong, S. Han, B. Bae, and S. Jeon, "Flexible protective film: Ultrahard, yet flexible hybrid nanocomposite reinforced by 3D inorganic nanoshell structures," *Advanced Functional Materials*, vol. 31, p. 2 010 254, 2021. JCR Q1, 95.2% IF 19.9

1. **D. Park**, Y. Choi, and J.-W. Hong, "Numerical study on seismic behavior of a three-story RC shear wall structure," *Journal of Earthquake Engineering Society of Korea*, vol. 25, no. 3, pp. 111–119, 2021, KCI.

Conference Presentations

26. Y. Kim, **D. Park**, and J.-W. Hong, "Location-dependent damage mapping and fragility analysis for enhanced blast safety assessment of 2D concrete slab," in *KSCM Conference 2026*, Poster, Daejeon, Korea, Feb. 2026.

25. H. Lee, **D. Park**, and J.-W. Hong, "Optimization-driven failure analysis of concrete cylinders under compression using the continuous surface cap model," in *KSCM Conference 2026*, Poster, Daejeon, Korea, Feb. 2026.

24. M. Kim, **D. Park**, and J.-W. Hong, "Simulations of debris flow impact on a transmission tower reinforced with a wire system," in *COSEIK Academic Symposium*, Busan, Korea, Apr. 2025.

23. H. Lee, **D. Park**, and J.-W. Hong, "Parametric study on cylindrical concrete compression using the continuous surface cap model," in *The Thirty-sixth KKHTCNN Symposium on Civil Engineering*, Hong Kong, China, Dec. 2025.

22. T. Lee, **D. Park**, Y. Lee, and J.-W. Hong, "Machine learning-based surrogate modeling of blast parameters for cuboid explosive charges," in *The 4th International Conference on COMPSAFE2025*, Kobe, Japan, Jul. 2025.

21. **D. Park**, T. Lee, and J.-W. Hong, "Blast damage mechanism of RC slabs using the continuous surface cap model (CSCM)," in *KSCE 2025 Convention*, Osong, Korea, Nov. 2025.

20. M. Kim, W. Jang, **D. Park**, and J.-W. Hong, "Fragility analyses of transmission towers under debris flows," in *KSCE 2024 Convention*, Poster, Jeju, Korea, Oct. 2024.

19. M. Kim, **D. Park**, and J.-W. Hong, "Finite element analyses of a transmission tower subjected to debris flow impact," in *The Thirty-fifth KKHTCNN Symposium on Civil Engineering*, Shanghai, China, Nov. 2024.

18. T. Lee, **D. Park**, Y. Choi, Y. Lee, and J.-W. Hong, "Blast responses of a reinforced concrete slab using the arbitrary Lagrangian-Eulerian method," in *The Fifteenth International Conference on Computational Structures Technology (CST)*, Prague, Czech Republic, Sep. 2024.

17. T. Lee, **D. Park**, Y. Lee, and J.-W. Hong, "Machine learning-based surrogate model for predicting blast parameters from cuboid charge," in *KSCM Conference 2024*, Poster, Seoul, Korea, Dec. 2024.

16. **D. Park**, T. Lee, Y. Lee, and J.-W. Hong, "Failure analyses of a reinforced concrete slab under blast loading," in *The Thirty-fifth KKHTCNN Symposium on Civil Engineering*, Shanghai, China, Nov. 2024.

15. H. Kim, Y. Choi, **D. Park**, M. Lee, G. Lee, B. Kim, and J.-W. Hong, "High-fidelity finite element model for acoustic emission sensors," in *KSCM Conference 2023*, Poster, Gangneung, Korea, Jun. 2023.

14. H. Kim, **D. Park**, M. Lee, G. Lee, B. Kim, and J.-W. Hong, "Numerical analyses of piezoelectric element-based stacked accelerometers," in *COSEIK Academic Symposium*, Chuncheon, Korea, Nov. 2023.

13 **D. Park**, Y. Choi, S. Kim, S. Jin, and J.-W. Hong, "Numerical model of a masonry wall subjected to earthquakes using the continuous surface cap model (CSCM)," in *KSCE 2023 Convention*, Yeosu, Korea, Oct. 2023.

12 **D. Park**, T. Lee, Y. Lee, Y. Choi, and J.-W. Hong, "Numerical analysis of a reinforced concrete slab subjected to blast loading," in *The Thirty-fourth KKHTCNN Symposium on Civil Engineering*, Pattaya, Thailand, Nov. 2023.

11 W. Jang, S. Kim, Y. Choi, **D. Park**, and J.-W. Hong, "Signal filter design for acceleration to measure displacements of masonry walls," in *The Thirty-third KKHTCNN Symposium on Civil Engineering*, Queenstown, Singapore, Nov. 2022.

10 H. Kim, **D. Park**, Y. Choi, and J.-W. Hong, "Geometry optimization of resonance acoustic emission sensors," in *The Thirty-third KKHTCNN Symposium on Civil Engineering*, Queenstown, Singapore, Nov. 2022.

9 **D. Park**, Y. Choi, S. Kim, and J.-W. Hong, "Failure analysis of a U-shaped masonry wall via shaking table tests," in *The Thirty-third KKHTCNN Symposium on Civil Engineering*, Queenstown, Singapore, Nov. 2022.

8 **D. Park**, Y. Choi, S. Kim, and J.-W. Hong, "Seismic behavior and analysis of a U-shaped masonry fence wall," in *Earthquake Engineering Society of Korea (EESK) Annual Conference*, Sep. 2022.

7 **D. Park**, Y. Choi, S. Lee, B. Kim, G. Lee, M. Lee, and J.-W. Hong, "Design optimization of acoustic emission sensor," in *KSCM Annual Conference*, Poster, PyeongChang, Korea, Jun. 2022.

6 Y. Hwang, **D. Park**, and J.-W. Hong, "Fracture analyses using a coupling approach of Voronoi-cell lattice model and finite element method," in *SPIE Smart Structures/NDE Digital Forum*, Mar. 2021.

5 **D. Park**, Y. Choi, and J.-W. Hong, "Seismic response analysis of a three-story RC shear wall structure," in *Earthquake Engineering Society of Korea (EESK) Annual Conference*, Mar. 2021.

4 **D. Park**, Y. Choi, H. Jang, and J.-W. Hong, "Seismic assessment of cracked masonry structures by shaking table tests," in *SPIE Smart Structures/NDE Digital Forum*, Mar. 2021.

3 Y. Choi, **D. Park**, H. Jang, and J.-W. Hong, "Seismic performance evaluation of cracked masonry structures via shaking table tests," in *Earthquake Engineering Society of Korea (EESK) Annual Conference*, Sep. 2020.

2 **D. Park**, Y. Choi, and J.-W. Hong, "Evaluation of floor response spectrum of RC shear wall structure with high fidelity FE model," in *Earthquake Engineering Society of Korea (EESK) Annual Conference*, Sep. 2020.

1 H. Jang, Y. Choi, **D. Park**, and J.-W. Hong, "Seismic risk analysis of deteriorated structures with high-fidelity FEM model," in *The Thirty-second KKHTCNN Symposium on Civil Engineering*, Daejeon, Korea, Oct. 2019.

Research Projects

Principal investigator

2026 – 2029 Development of AI-Based Automated Material Model Calibration and Structural Analysis Techniques for High-Fidelity Numerical Simulation. PI: Dawon Park. Basic Research Program (Type B), National Research Foundation of Korea (RS-2026-25558353).

Intellectual Property

Patents

4. J.-W. Hong, **D. Park**, and M. Kim, "Ground support structure for transmission tower," Korean Patent Application 10-2025-0153146, 2025.
3. J.-W. Hong, **D. Park**, and M. Kim, "Structural monitoring system and method based on risk assessment," Korean Patent Application 10-2025-0160980, 2025.
2. J.-W. Hong, **D. Park**, M. Kim, and W. Jang, "Structural reinforcement system for transmission towers and construction method thereof," Korean Patent Application 10-2025-0001455, 2025.
1. J.-W. Hong, **D. Park**, M. Kim, and S. Kim, "Debris flow drop test apparatus," Korean Patent Application 10-2025-0085354, 2025.

Software Registration

4. J.-W. Hong, Y. Choi, **D. Park**, S. Kim, S. Lee, S. Ok, D. Kim, and Y. Jeong, "Explosive source detection model in panoramic images," Registration number: C-2022-015029, 30 March 2022, 2022.
3. J.-W. Hong, Y. Choi, **D. Park**, S. Kim, S. Lee, S. Ok, D. Kim, and Y. Jeong, "Gas pipe housing detection model in panoramic images," Registration number: C-2022-015030, 30 March 2022, 2022.
2. J.-W. Hong, Y. Choi, **D. Park**, S. Kim, S. Lee, S. Ok, D. Kim, and Y. Jeong, "Gas tank detection model in panoramic images," Registration number: C-2022-015031, 30 March 2022, 2022.
1. J.-W. Hong, Y. Choi, **D. Park**, S. Kim, S. Lee, S. Ok, D. Kim, and Y. Jeong, "Outdoor unit detection model in panoramic images," Registration number: C-2022-015032, 30 March 2022, 2022.